Synthetic Lecture: Foundations of Algorithms

Section 1: Asymptotic Complexity

Big-O notation describes the upper bound of an algorithm's execution time.

For example, binary search runs in O(log n) time, while linear search requires O(n).

Section 2: Graph Traversal and Shortest Paths

Breadth-First Search (BFS) utilizes a First-In-First-Out (FIFO) queue.

Depth-First Search (DFS) relies on recursion or an explicit stack.

Dijkstra's algorithm computes the shortest path on non-negative weighted graphs.